EXP NO-4
DATE-

SETUP AND CONFIGURE A LAN (Local Area Network) USING A SWITCH AND ETHERNET CABLES IN YOUR LAB

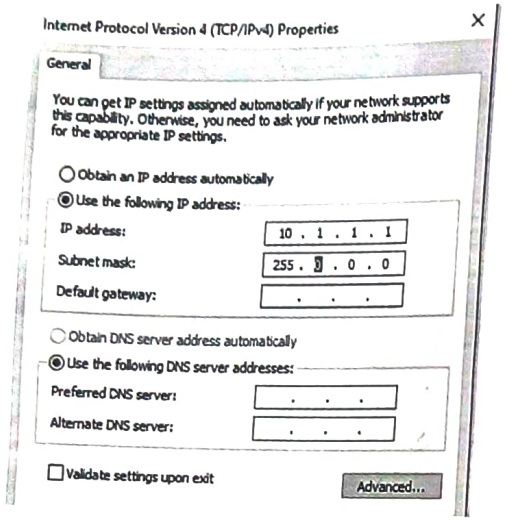
AIM: Setup and Configure a LAN (Local Area Network) using a Switch and Ethernet Cables in your Lab.

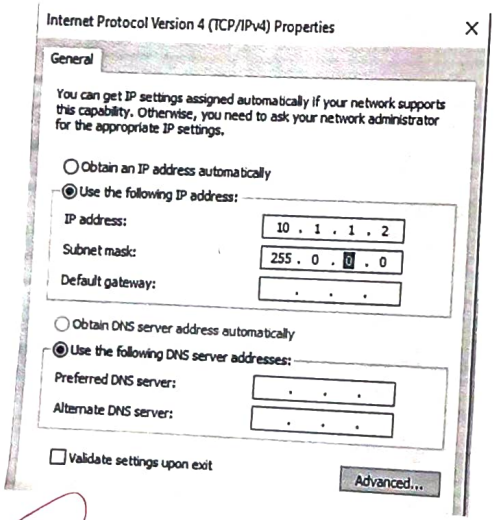
What is a LAN?

A Local Area Network (LAN) refers to a network that connects device within a limited area, such as an office building, school or home. It enables users to share resources, including data, printers and internet access. A local area network (LAN) switch servers as the primary connecting devices, managing and directing communications within the local network. Each connected device and LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

How to setup a LAN?

1. Plan and Design an appropriate network topology taking into account network requirement and equipment location.
2. You can take 4 computers, switch with 8, 16, (or) 24 ports which is sufficient for networks of the sizes, and 4 Ethernet cables.
3. Connect your computers to the network switch via an Ethernet cable, which is as simple as plugging one end of the Ethernet cable into your computer and the other end





into your network switch.

4) Assign IP address to your PC's.

(i) Log on the client computer as Administrator (or) as Owner.

(ii) Click network and Internet Connections.

(iii) Right click local Area Connection/ Ethernet → Go to properties → Select Internet Protocol (TCP/IPV4) → Click on properties → Select use the following ip address option and assign ip address.

similarly assign IP address to all the PC's connectioned to switch.

eg:-

PC1 - IP address: 10.1.1.1, subnet mask 255.0.0.0.

5) Configure a network switch:

1. Connect your computer to the switch: Connect your computer switch using an Ethernet cable.
2. Log in to the web interface: Open a web-browser, and enter the IP address of the switch, Enter username, password and login.
3. Configure basic settings: On you're logged in, you will able to configure basic setting for the switch.
4. Assign IP address as: 10.1.1.5, subnet mask 255.0.0.0.

6) Check the connectivity between switch and other machine by using ping command in the command prompt of the device.

7) Select a folder, → go to properties → Click sharing tab → share it with everyone on the same LAN.

8) Try to access the shared folder from others computers of the network.

RESULT: Thus, the setup and configure a LAN using a switch and Ethernet cables has been implemented and executed successfully.